Soon smartphones will identify deepfakes even before they are made

Deepfakes are one of the most critical problems being faced by the digital world. They not only spread mistrust, but they also help in expanding the outreach of fake news. But now, a new company has devised an ingenious method that can be used for identifying deepfakes right at the beginning, that is, when an individual is capturing images and videos for the same.

The company's technology provides a new 'secure' capture mode in the device's native camera app (Credit: Pixabay)

Truepic has developed a technology that is embedded in a smartphone. This technology provides a new 'secure' capture mode in the device's native camera app. Engaging this mode and pressing the capture button produces a digital photo that contains cryptographically-sealed provenance data, formatted in accordance with emerging open standards. The data allows a recipient to authenticate the pixels, date and time, geolocation, and 3D depth map that were securely-acquired and sealed into the file at the time of capture.

The company has partnered with Qualcomm, and it has embedded its technology called Truepic Foresight in the company's Snapdragon 865 5G Mobile Platform. The technology takes advantage of the chipset's underlying hardware security.

Scientists develop 'mini-brains' to help robots recognise pain and self-repair

Using a brain-inspired approach, scientists from Nanyang Technological University, Singapore (NTU Singapore), have developed a way for robots to recognise pain and self-repair, when damaged, using artificial intelligence (AI).

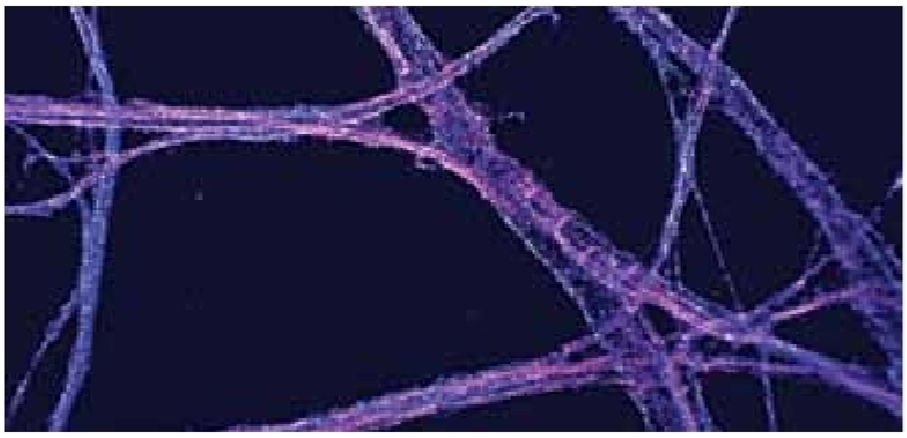
The system has AI-enabled sensor nodes to process and respond to 'pain' arising from pressure exerted by a physical force. The system also allows the robot to detect and repair its own damage when minorly 'injured,' without the need for human intervention. Currently, robots use a network of sensors to generate information about their immediate environment.

The new NTU approach embeds AI into the network of sensor nodes, connected to multiple small and less-powerful processing units that act like 'mini-brains' distributed on the robotic skin. This means learning happens locally, and the wiring requirements and response time for the robot are reduced five to ten times compared to conventional robots, say the scientists.

Combining the system with a type of self-healing ion gel material means that the robots, when damaged, can recover their mechanical functions without human intervention.

The system uses a type of self-healing ion gel material that enables the robot to restore its mechanical functions (Credit: NTU Singapore)

New virtual reality software allows scientists to 'walk' inside cells



Virtual reality software, which allows researchers to 'walk' inside and analyse individual cells, could be used to understand fundamental problems in biology and develop new treatments for disease

The software, called vLUME, was created by scientists at the University of Cambridge and 3D image analysis software company Lume VR Ltd. It allows super-resolution microscopy data to be visualised and analysed in virtual reality and to be used to study everything from individual proteins to entire cells.

Super-resolution microscopy, which was awarded the Nobel Prize for Chemistry in 2014, makes it possible to obtain images at the nanoscale by using clever tricks of physics to get around the limits imposed by light diffraction. This has allowed researchers to observe molecular processes as they happen. However, a problem has been the lack of ways to visualise and analyse this data in three dimensions.